BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: MRD-11TH 190623(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-11TH 190623(A)
VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. The percentage uncertainty in the measurement of (3.76 ± 0.25)m is:

(A) 4%
(B) 6.6%
(C) 25%
(D) 3.76%

Q2. The precision of the measurement 75N is:

(A) 7.5N
(B) 75N
(C) 1N
(D) 10N

Q3. Example of vector product is:

(A) Angular momentum
(B) Work
(C) Power
(D) None of these

Q4. Two or more concurrent forces can be balanced by a single force called.

(A) Resultant force
(B) Equilibrant force
(C) Conservative force
(D) Deforming force

Q5. The time rate of change of linear momentum of an object is called:

(A) Impulse
(B) Power
(C) Torque
(D) Force

Q6. For an ideal projectile, through out its flight, the horizontal component of velocity

(A) Changes continuously
(B) Is minimum at highest point
(C) Remains constant
(D) Is maximum at highest point

Q7. The efficiency of fluorescent lamp is:

(A) 5%
(B) 10%
(C) 15%
(D) 20%

Q8. When the speed of your car is doubled, by what factor does its kinetic energy increases?

(A) 1
(B) 2
(C) 4
(D) 8

Q9. For a hoop, rolling down the inclined plane of height h, the expression for velocity at the bottom is:

(A) $\sqrt{gh}$
(B) $\sqrt{2gh}$
(C) $\sqrt{(4/3\ gh)}$
(D) $2gh$

Q10. The time taken by geostationary satellite to orbit the earth is:

(A) 365 days
(B) 30 days
(C) 24 hours
(D) 1 hour

Q11. Venture meter is a device used to measure the:

(A) Mass of fluid
(B) Viscosity of fluid
(C) Speed of fluid
(D) Density of fluid

Q12. The SI unit of spring constant is:

(A) Kgm
(B) kg/s
(C) Kg/m
(D) kg/s^2

Q13. To make the frequency double of a spring oscillation, we haave to:

(A) Reduce the mass to one fourth
(B) Quadruple the mass
(C) Double the mass
(D) Half the mass

Q14. At one atmospheric pressure and 0C° temperature speed of sound in air is 332 ms^{-1}. The speed of sound in air at two atmospheric pressure and 0C temperature will be:

(A) $332ms^{-1}$
(B) $340ms^{-1}$
(C) $664ms^{-1}$
(D) $350ms^{-1}$

Q15. If the amplitude of a wave is doubled, then its intensity is:

(A) Doubled
(B) Halved
(C) Quadrupled
(D) One fourth

Q16. For glass of refractive index 1.55, the angle of polarization is:

(A) 55°
(B) 54°
(C) 57°
(D) 53°

Q17. The SI unit of molar specific heat is:

(A) $Jkg^{-1}k^{-1}$
(B) $Jmol^{-1}k^{-1}$
(C) $Jmolk^{-1}$
(D) $Jkg^{-1}k$

Q18. A real gas can be approximated to an ideal gas at:

(A) Low density
(B) High pressure
(C) High density
(D) Low temperature

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 40

Q2. Write short answers to any ten (10) of the following parts. All parts carry equal marks:-

(i) Explain with examples, the concept of precision and accuracy.

(ii) $\vec{A}$ and $\vec{B}$ are two non-zero vectors. How can their scalar product be zero? And how can their vector product be zero.

(iii) Explain the warning "Never use a large wrench to tighten a small bolt".

(iv) The time rate of change of linear momentum is called force. Discuss.

(v) Calculate the angle of projection for which kinetic energy at the summit is equal to one-fourth of its kinetic energy at point of projection.

(vi) Is kinetic energy a vector quantity? Explain.

(vii) What should be the orbital speed to launch a satellite in a circular orbit 900km above the surface of the earth?

(viii) Why does smoke rise faster in a chimney on a windy day?

(ix) Give one practical example each of free and forced oscillation.

(x) Calculate the length of a second pendulum having time period 2S at a place where g = $9.8ms^{-2}$.

(xi) Why does a sound wave travel faster in solid than in gases?

(xii) Explain any two applications of polarized light.

(xiii) Explain the energy ratios of refrigerator.

Section – C

Maximum Marks: 27

Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(i) For an ideal projectile, derive expressions of its maximum height and time of flight.

(ii) Show that for two complementary angles of measure 75° and 15° range of a projectile is same.

Q4.

(i) Explain the concept of apparent weight?

(ii) A 70kg man is standing on a scale in an elevator which is accelerating, as it heads for the top floor of a building at $4ms^{-1}$. What apparent weight will show on the scale?

Q5.

(i) What is simple pendulum? Derive a relation for its time period and show that time period is independent of mass of the bob.

(ii) Calculate the frequency of a simple pendulum if its length is "1m".

Q6.

(i) Explain any two (2) applications of first law of thermodynamics.

(ii) In a certain process, 400J of heat energy is supplied to a system and at the same time 150J of work is done by the system. What is the increase in internal energy?